

PixelUnshuffle#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.pixelshuffle.PixelUnshuffle.html>

Description:

Reverse the PixelShuffle operation. Reverses the PixelShuffle operation by rearranging elements in a tensor of shape $(*, C, H \times r, W \times r)$ to a tensor of shape $(*, C \times r^2, H, W)$, where r is a downscale factor. See the paper: Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network by Shi et al. (2016) for more details.

`downscale_factor` (int) – factor to decrease spatial resolution by

Input: $(*, C_{in}, H_{in}, W_{in})$, where $*$ is zero or more batch dimensions

Function Signature

```
class  
torch.nn.modules.pixelshuffle.PixelUnshuffle(downscale_factor)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> pixel_unshuffle = nn.PixelUnshuffle(3)  
>>> input = torch.randn(1, 1, 12, 12)  
>>> output = pixel_unshuffle(input)  
>>> print(output.size())  
torch.Size([1, 9, 4, 4])
```

Detailed Documentation

Documentation

Reverse the PixelShuffle operation.

Reverses the PixelShuffle operation by rearranging elements in a tensor of shape $(*, C, H \times r, W \times r)$ to a tensor of shape $(*, C \times r^2, H, W)$, where r is a downscale factor.

See the paper: Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network by Shi et al. (2016) for more details.

`downscale_factor` (int) – factor to decrease spatial resolution by

Input: $(*, C_{in}, H_{in}, W_{in})$, where $*$ is zero or more batch dimensions

Output: $(*, C_{out}, H_{out}, W_{out})$, where

Return the extra representation of the module.

Runs the forward pass.